

Appendix 19

Completeness of the bosonic basis set for the Luttinger model

In this appendix we prove the completeness of the *extended* bosonic basis set $|N_L, N_R, \{\mathcal{N}_q\}\rangle$, where N_L and N_R are the numbers of left and right movers counted from the reference “vacuum” in which all the right-mover states are occupied for $k < k_F$ and all the left-mover states are occupied for $k > -k_F$. Thus N_L and N_R can be positive or negative, with a positive value indicating an excess of particles (relative to the “vacuum”) and a negative value indicating a deficiency of particles. The basis set (9.37) defined in Section 9.4 is just a special case with $N_L = N_R = 0$. The energies of these states (relative to the energy of the “vacuum”) are

$$E[N_L, N_R, \{\mathcal{N}_q\}] = \hbar v_F \left[\frac{\pi}{L} (N_L^2 + N_R^2) + \sum_{q \neq 0} |q| \mathcal{N}_q \right], \quad (\text{A19.1})$$

where the first term $\hbar v_F \frac{\pi}{L} (N_L^2 + N_R^2)$ represents the excess ground-state energy of the non-interacting system with N_τ extra particles and no boson excitations (Exercise 1).¹

The grand-canonical partition function² $\mathcal{Z} = \text{Tr}\{e^{-\beta \hat{H}_0}\}$ can be calculated in both the bosonic and the fermionic basis sets. The sum over N_τ , being rapidly convergent, can be safely extended to $-\infty$. In the bosonic basis set we have

$$\begin{aligned} \mathcal{Z}_B &= \sum_{N_R=-\infty}^{\infty} e^{-\beta \hbar \pi v_F \frac{N_R^2}{L}} \times \sum_{N_L=-\infty}^{\infty} e^{-\beta \hbar \pi v_F \frac{N_L^2}{L}} \\ &\times \prod_{q \neq 0} \sum_{\mathcal{N}_q=0}^{\infty} e^{-\beta \hbar v_F |q| \mathcal{N}_q}. \end{aligned} \quad (\text{A19.2})$$

Setting $w \equiv e^{-\frac{\beta \hbar \pi v_F}{L}}$ and carrying out the sum of the geometric series on the last line of Eq. (A19.2) we get

$$\begin{aligned} \mathcal{Z}_B &= \left(\sum_{n=-\infty}^{\infty} w^{n^2} \right)^2 \times \prod_{q \neq 0} \frac{1}{1 - w^{\frac{qL}{\pi}}} \\ &= \left(\sum_{n=-\infty}^{\infty} w^{n^2} \right)^2 \times \left(\prod_{n=1}^{\infty} \frac{1}{1 - w^{2n}} \right)^2, \end{aligned} \quad (\text{A19.3})$$

¹ The reason for the quadratic behavior is that the energy of an extra particle vanishes at the Fermi level.

² Notice that the chemical potential term is built into our definition of $\hat{H}_0$.

where we have used the fact that $q = \frac{2\pi}{L}n$, with n a nonzero integer.

On the other hand, in the fermionic basis set, we can write

$$\mathcal{Z}_F = \prod_k \left(\sum_{n_{k,R}=0}^1 e^{-\beta \hbar v_F (k-k_F)(n_{k,R}-n_{k,R}^{(0)})} \times \sum_{n_{k,L}=0}^1 e^{+\beta \hbar v_F (k+k_F)(n_{k,L}-n_{k,L}^{(0)})} \right), \quad (\text{A19.4})$$

with $n_{k,\tau}^{(0)}$ defined in Eq. (9.1). Carrying out the sums over fermionic occupation numbers (0 or 1) we get

$$\begin{aligned} \mathcal{Z}_F &= \prod_{k>k_F} \left(1 + w^{\frac{(k-k_F)L}{\pi}} \right) \times \prod_{k<k_F} \left(1 + w^{-\frac{(k-k_F)L}{\pi}} \right) \\ &\quad \times \prod_{k<-k_F} \left(1 + w^{\frac{(k+k_F)L}{\pi}} \right) \times \prod_{k>-k_F} \left(1 + w^{-\frac{(k+k_F)L}{\pi}} \right). \end{aligned} \quad (\text{A19.5})$$

Now setting $k \pm k_F = \frac{2\pi}{L}n$ we rewrite the above equation as

$$\mathcal{Z}_F = \left(\prod_{n=1}^{\infty} (1 + w^{2n-1})^2 \right)^2. \quad (\text{A19.6})$$

To show that Eqs. (A19.3) and (A19.6) are indeed two different representations of the same quantity we refer the reader to Eqs. (8.180.3) and (8.181.2) of Gradshteyn–Ryzhik (1965, p. 921), which give two alternative representations of the Jacobi elliptic theta-function

$$\begin{aligned} \theta_3(0, w) &= \sum_{n=-\infty}^{\infty} w^{n^2} \\ &= \prod_{n=1}^{\infty} (1 + w^{2n-1})^2 (1 - w^{2n}), \quad (|w| < 1). \end{aligned} \quad (\text{A19.7})$$

Then we see that $\mathcal{Z}_B = \mathcal{Z}_F$ and by the argument of Section 9.4 we conclude that the fermionic and bosonic basis sets are equivalent.